Repayment Resilience | Product Case Study

Independent portfolio project | Tina Liang | November 2026 | Simulated data only

Problem and idea

Some borrowers may miss payments because their due date falls before payday rather than because they can't pay. This case tests whether aligning due dates with income timing improves repayment without increasing hardship, and sends people in real hardship to a person for an affordability review.

Phase 1 and Phase 2

Phase 1: eligibility check, due-date preview, consent, audit trail, payment event tracking, manual hardship review and a quick off switch. No new credit. Phase 2: hardship or settlement lending, with affordability review, underwriting, clear terms and sign-off from Credit Risk, Finance, Legal, Compliance, any partner bank and Engineering.

Test design

Randomize by customer, 50/50, analyze everyone as assigned. Decision measure: first payment and dollars collected by day 60 after assignment, on a fixed calendar, because moving a due date also moves the finish line for on-time. Supporting: on time vs own due date, collected by day 30. Stop signals: complaints, NSF fees, later missed payments, accounts rolling further behind. Sample size, planned on the decision measure (90.4% baseline, 5% significance, 80% power): 3,085 per group for a 2-point lift, 1,293 for 3, 404 for 5. The demo (about 1,700 per group) can detect 3 points but not 2.

What the simulation shows (not a forecast)

On time vs own due date: 80.6% aligned vs 72.0% control, +8.6 pts (95% CI 5.8 to 11.5), but I built a 7.5-point effect into the data. On a fixed calendar the gain mostly goes away: collected by day 30 -1.4 pts (-4.3 to 1.4), by day 60 +1.6 pts (-0.3 to 3.5), or +$6.7 per customer (-$0.3 to $13.8). The on-time gain doesn't turn into a statistically clear gain in day-60 collections, so on these results I'd keep testing rather than roll out. 179 accounts would go to manual hardship review.

Settlements and on-load collections

Offer a settlement only if expected net recovery beats normal collections after costs, and the customer can afford it. It only pays off if completed, so I track acceptance (out of offers), completion (out of acceptances) and overall recovery separately. With costs split by timing (1% setup on acceptance, 2% on completion, 2% extra if it fails), a 50% settlement against 25% expected collections needs more than about 27% of accepted settlements to complete; charging cost only on completion would understate that at 19%. On-load collections can lift recovery but can leave people short for essentials: opt-in, capped, with exemptions and complaint and hardship guardrails.

Decision

Don't roll out yet. Keep testing, sized on day-60 collections (about 1,300 customers per group for a 3-point lift), with customer-protection guardrails in place.

Money and go/no-go

Losses avoided = (control charge-off roll rate - treatment charge-off roll rate) x balance at risk x loss rate at charge-off. No dollar estimate without real charge-off data. Scale only with a credible lift on the day-60 collection measure, no meaningful harm, and legal and risk approval.

Code

src/analysis.py, src/power_analysis.py, src/settlement_breakeven.py, 15 automated tests, a SQL template for cure and roll rates with a standalone month-gap check, and the PRD. github.com/tinabigbig/repayment-resilience-case-study